



ECEN 314 - Signals and Systems

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Module 12 - Fourier Transforms – Part 2

Learning Objectives



- Be able to find Fourier transforms of signals that contain delta functions in either the time domain or the frequency domain.
- Understand the difference between a power and an energy signal.
- Be able to compute the power/energy of a signal directly or using Parseval's theorem.
- Understand the various definitions of bandwidth and be able to compute the bandwidth of typical signals.

Fourier Transform of Delta Functions



When taking Fourier transforms involving delta functions, it is helpful to use the sifting property of delta functions:

$$\int_a^b \delta(t - t_o) x(t) dt = \begin{cases} x(t_o), & a < t_o < b, \\ 0, & t_o < a, \text{ or } t_o > b. \end{cases}$$

Consider taking the Fourier Transform of the delta function

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi ft} dt = 1.$$

Likewise, with the inverse Fourier Transform,

$$\mathcal{F}^{-1}[\delta(f)] = \int_{-\infty}^{\infty} \delta(f) e^{j2\pi ft} df = 1.$$

Therefore, if $x(t) = \delta(t)$ is input to a circuit/system with transfer function $H(f)$, then the output will be $Y(f) = H(f)X(f) = H(f)$ (since $X(f) = 1$ in this case) so that $y(t) = \mathcal{F}^{-1}[H(f)] = h(t)$. This is why $h(t)$ is called the impulse response as it is the output that appears when the input is a unit impulse function.

Fourier Transforms of Sinusoids



Starting with $\mathcal{F}^{-1}[\delta(f)] = 1$, and then using the frequency shifting property:

$$\mathcal{F}^{-1}[\delta(f - f_o)] = e^{-j2\pi f_o t},$$

$$\mathcal{F}^{-1}[\delta(f + f_o)] = e^{j2\pi f_o t},$$

and then the linearity property,

$$\mathcal{F}^{-1}\left[\frac{1}{2}\delta(f + f_o) + \frac{1}{2}\delta(f - f_o)\right] = \frac{1}{2}e^{j2\pi f_o t} + \frac{1}{2}e^{-j2\pi f_o t} = \cos(2\pi f_o t),$$

$$\mathcal{F}^{-1}\left[\frac{1}{2j}\delta(f + f_o) - \frac{1}{2j}\delta(f - f_o)\right] = \frac{1}{2j}e^{j2\pi f_o t} - \frac{1}{2j}e^{-j2\pi f_o t} = \sin(2\pi f_o t).$$

Therefore,

$$\mathcal{F}[\cos(2\pi f_o t)] = \frac{1}{2}\delta(f + f_o) + \frac{1}{2}\delta(f - f_o),$$

$$\mathcal{F}[\sin(2\pi f_o t)] = \frac{j}{2}\delta(f - f_o) - \frac{j}{2}\delta(f + f_o).$$

Not surprisingly, all the frequency content of sinusoids is contained at $f = \pm f_o$.

Fourier Transforms of Periodic Signals



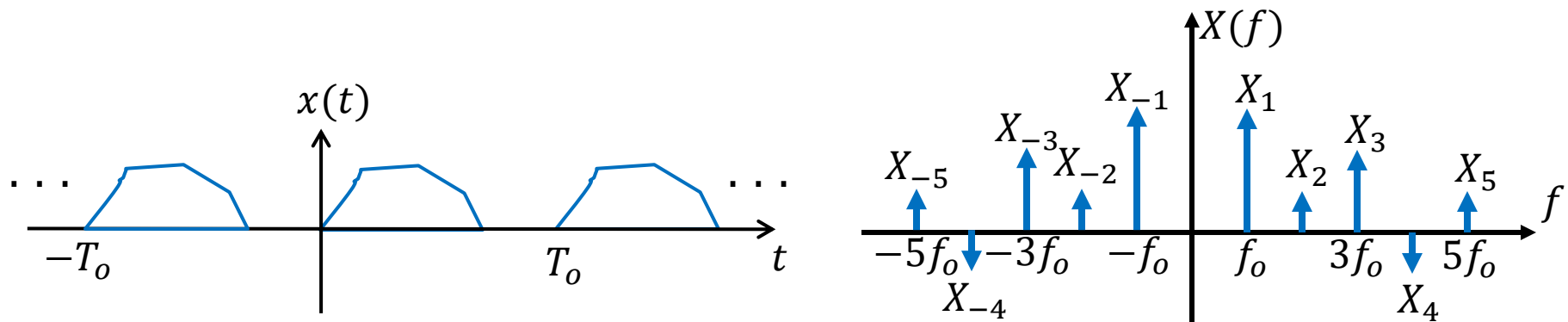
Suppose $x(t)$ is a periodic signal with a period of T_o (fundamental frequency of $f_o = 1/T_o$) with some Fourier series representation

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j2\pi k f_o t}.$$

Taking F.T.'s of both sides of the equation,

$$\mathcal{F}[x(t)] = \mathcal{F}\left[\sum_{k=-\infty}^{\infty} X_k e^{j2\pi k f_o t}\right] = \sum_{k=-\infty}^{\infty} X_k \mathcal{F}[e^{j2\pi k f_o t}] = \sum_{k=-\infty}^{\infty} X_k \delta(f - k f_o).$$

Periodic signals have a Fourier transform consisting of discrete components (not a continuum of frequency components). We refer to this as a *line spectrum*.

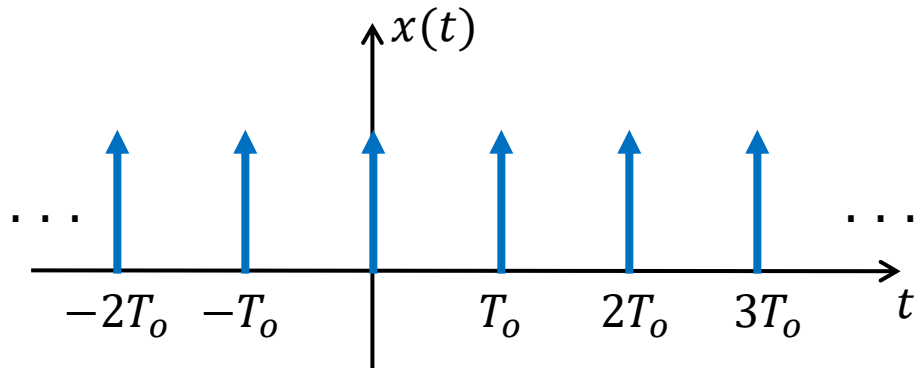


The Impulse Train



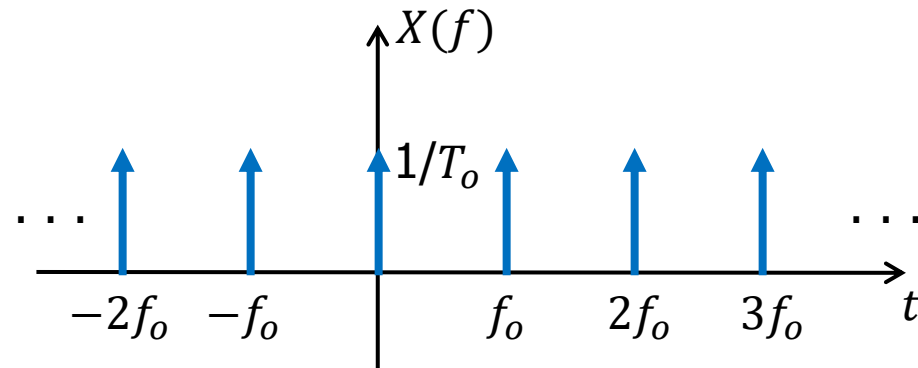
Later in the course we will use a periodic impulse train to convert signals from continuous time to discrete time.

$$x(t) = \sum_{m=-\infty}^{\infty} \delta(t - mt_o).$$



Since this signal is periodic, it has a Fourier Series representation with coefficients given by

$$\begin{aligned} X_k &= \frac{1}{T_o} \int_{-T_o/2}^{T_o/2} x(t) e^{-j2\pi k f_o t} dt \\ &= \frac{1}{T_o} \int_{-T_o/2}^{T_o/2} \delta(t) e^{-j2\pi k f_o t} dt = \frac{1}{T_o}. \end{aligned}$$



Therefore, the F.T. of the impulse train is itself an impulse train.

Fourier Transform of Unit Step



Start with the signum function,

$$\text{sgn}(t) = \begin{cases} 1, & t > 0, \\ -1, & t < 0. \end{cases}$$

It's derivative is $\frac{d}{dt}\{\text{sgn}(t)\} = 2\delta(t)$.

Therefore,

$$\frac{d}{dt}\{\text{sgn}(t)\} \leftrightarrow 2.$$

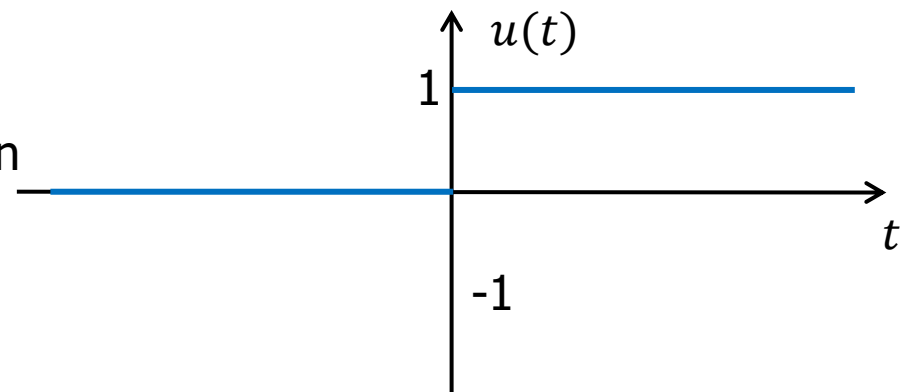
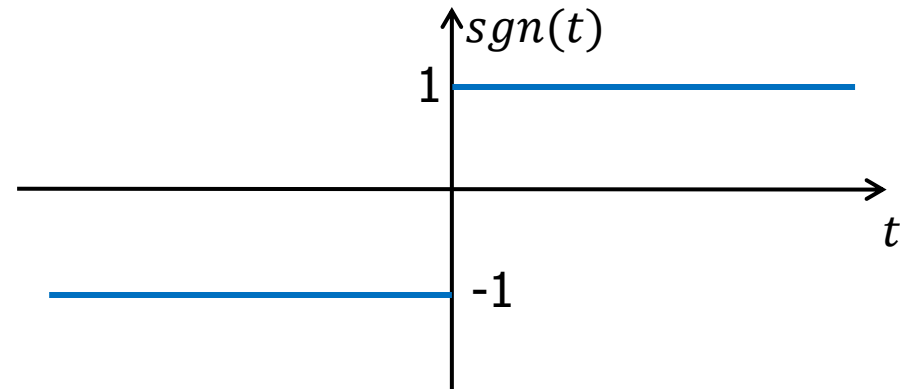
Using the integration property of F.T.'s, then

$$\text{sgn}(t) \leftrightarrow \frac{2}{j\omega} \text{ (or } \frac{1}{j\pi f}).$$

We can relate the unit step function to the signum function according to

$$u(t) = \frac{1}{2} + \frac{\text{sgn}(t)}{2}$$

$$\mathcal{F}[u(t)] = \frac{1}{2}\delta(\omega) + \frac{1}{j\omega} \text{ or } \frac{1}{2}\delta(f) + \frac{1}{j2\pi f}.$$



Note: The presence of the delta function in the spectrum of the unit step is the result of the step having a DC component (average value) of $\frac{1}{2}$.

Duality



- Another very useful property of Fourier transforms is called duality and involves the exchange of the roles of time and frequency in the Fourier transform relationship.

- Start with FT relationship: $x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df \Rightarrow x(t) \leftrightarrow X(f)$

- Negate time: $x(-t) = \int_{-\infty}^{\infty} X(f)e^{-j2\pi ft}df$

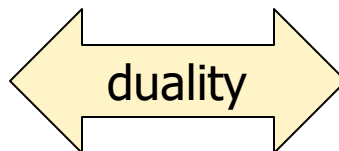
- Exchange roles of the time and frequency variables, $t \rightarrow f, f \rightarrow t$:

$$x(-f) = \int_{-\infty}^{\infty} X(t)e^{-j2\pi ft}dt \Rightarrow X(t) \leftrightarrow x(-f)$$

- In words, this means that if we know a certain Fourier transform pair, then we can exchange the roles of time and frequency to produce a new Fourier transform pair (to within a minus sign).

Example

$$\text{rect}(t) \leftrightarrow \text{sinc}(f)$$



$$\text{sinc}(t) \leftrightarrow \text{rect}(-f)$$

Fourier Transforms of Common Signals



- Based on the results of the last few slides, we can add several more signals to our table of Fourier transforms.
- This table together with the list from the previous module gives us a pretty good table of common signals.
- For more complicated signals, use these known signals together with various properties.

Signal	Transform, Hz, $X(f)$	Transform, rad/sec, $X(\omega)$
$\delta(t)$	1	1
1	$\delta(f)$	$2\pi\delta(\omega)$
$\cos(\omega_o t)$	$\frac{1}{2}\delta(f + f_o) + \frac{1}{2}\delta(f - f_o)$	$\frac{1}{2}\delta(\omega + \omega_o) + \frac{1}{2}\delta(\omega - \omega_o)$
$\sin(\omega_o t)$	$\frac{1}{2j}\delta(f + f_o) - \frac{1}{2j}\delta(f - f_o)$	$\frac{1}{2j}\delta(\omega + \omega_o) - \frac{1}{2j}\delta(\omega - \omega_o)$
$u(t)$	$\frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$	$\frac{1}{2}\delta(\omega) + \frac{1}{j\omega}$
$\text{sgn}(t)$	$\frac{1}{j\pi f}$	$\frac{2}{j\omega}$

Power/Energy in Signals

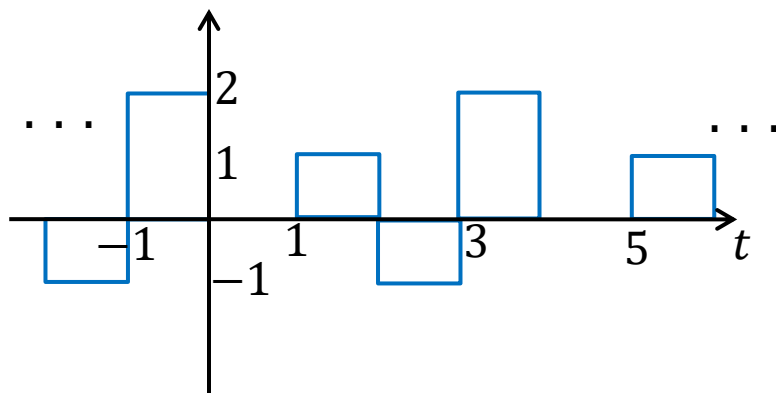


- Recall from your study of circuits, that if $x(t)$ is a voltage in a resistor, then its instantaneous power is given by $p(t) = x^2(t)/R$ while if it is a current then $p(t) = x^2(t) * R$.
- In order to allow us to study the power in a signal without having to worry about whether it is a current or a voltage, we often talk about a *normalized power* where it is assumed that $R = 1\Omega$, so that regardless of whether the signal is a voltage or a current, $p(t) = x^2(t)$.
- If we use the relationship between power and energy, $p(t) = \frac{dw}{dt}$, then in this context energy would be $w(t) = \int p(t)dt = \int x^2(t)dt$.
- To make these expressions not functions of time, we consider
 - ✓ Average Power = $P = \frac{1}{T_o} \int_{t_o}^{t_o+T_o} x^2(t)dt$ (for periodic signals)
 - ✓ Total Energy = $W = \int_{-\infty}^{\infty} x^2(t)dt$ (for aperiodic finite duration signals)
- Moving forward, we will just refer to these quantities as power and energy.

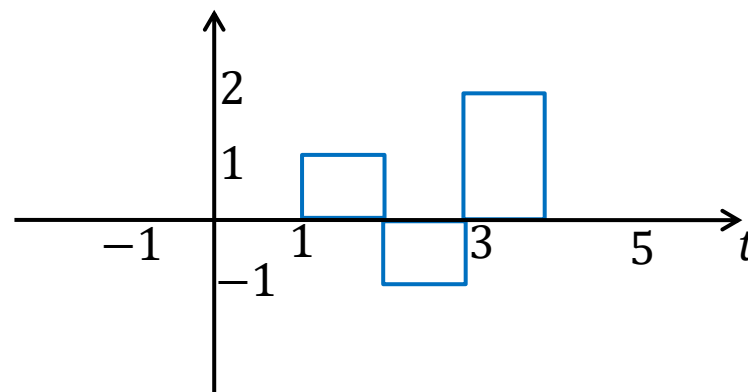
Power/Energy in Signals



- Generally, a signal that is periodic will have $P = \text{finite}$ but $W = \infty$. We refer to this as a *power type signal* as it is meaningful to talk about its power but not its energy.
- Generally, a signal that is aperiodic will have $P = 0$ but $W = \text{finite}$. We refer to this as an *energy type signal* as it is meaningful to talk about its energy but not its power.



A power type signal
 $P = 1.5, W = \infty$.



An energy type signal
 $P = 0, W = 6$.

Parseval's Theorem



- It turns out that we can determine the energy (or power) of a signal from its frequency content.

$$\begin{aligned} W &= \int_{-\infty}^{\infty} x^2(t) dt = \int_{-\infty}^{\infty} x(t) \left[\int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df \right] dt \\ &= \int_{-\infty}^{\infty} X(f) \left[\int_{-\infty}^{\infty} x(t) e^{j2\pi ft} dt \right] df = \int_{-\infty}^{\infty} X(f) X^*(f) df = \int_{-\infty}^{\infty} |X(f)|^2 df \end{aligned}$$

- This result is known as *Parseval's theorem* and it states that the energy of a signal is the same in the time domain and in the frequency domain

$$W = \int_{-\infty}^{\infty} x^2(t) dt = \int_{-\infty}^{\infty} |X(f)|^2 df \quad \text{or} \quad \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

- For power type signals, there is an equivalent result involving Fourier series coefficients

$$P = \frac{1}{T_o} \int_{t_o}^{t_o+T_o} x^2(t) dt = \sum_{k=-\infty}^{\infty} |X_k|^2$$

Example

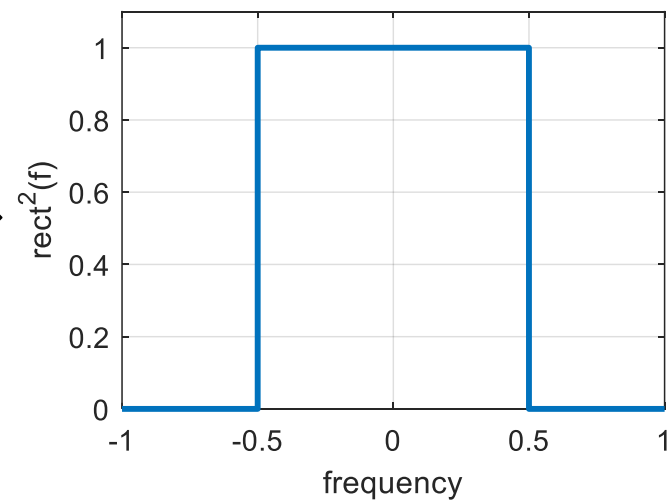
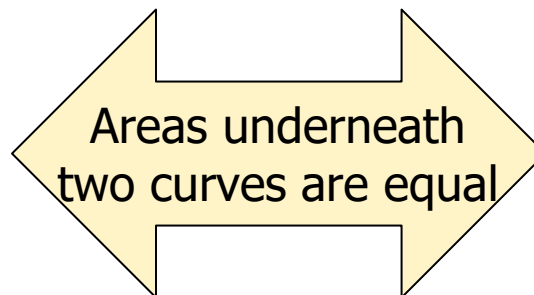
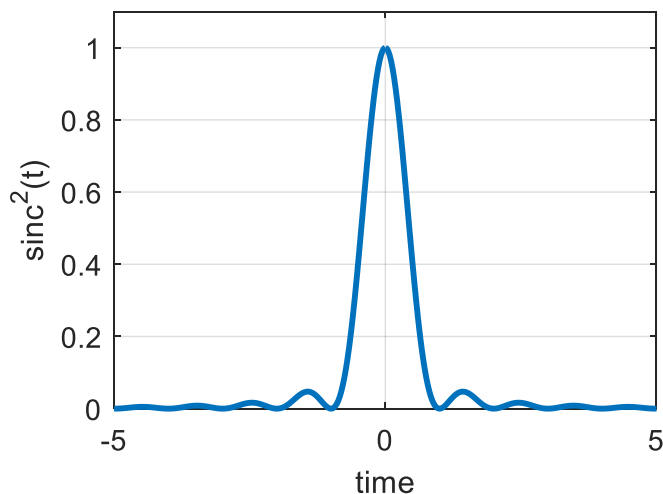


- Suppose we wanted to find the energy in the signal $x(t) = \text{sinc}(t)$. Performing the computation directly, we would need to calculate the integral

$$W = \int_{-\infty}^{\infty} \text{sinc}^2(t) dt = \int_{-\infty}^{\infty} \frac{\sin^2(\pi t)}{(\pi t)^2} dt.$$

- This integral would be rather tricky to do.
- Instead we can use Parseval's theorem together with the fact that $\text{sinc}(t) \leftrightarrow \text{rect}(f)$ so that

$$W = \int_{-\infty}^{\infty} \text{sinc}^2(t) dt = \int_{-\infty}^{\infty} \text{rect}^2(f) df = 1$$



Time/Frequency Relationships



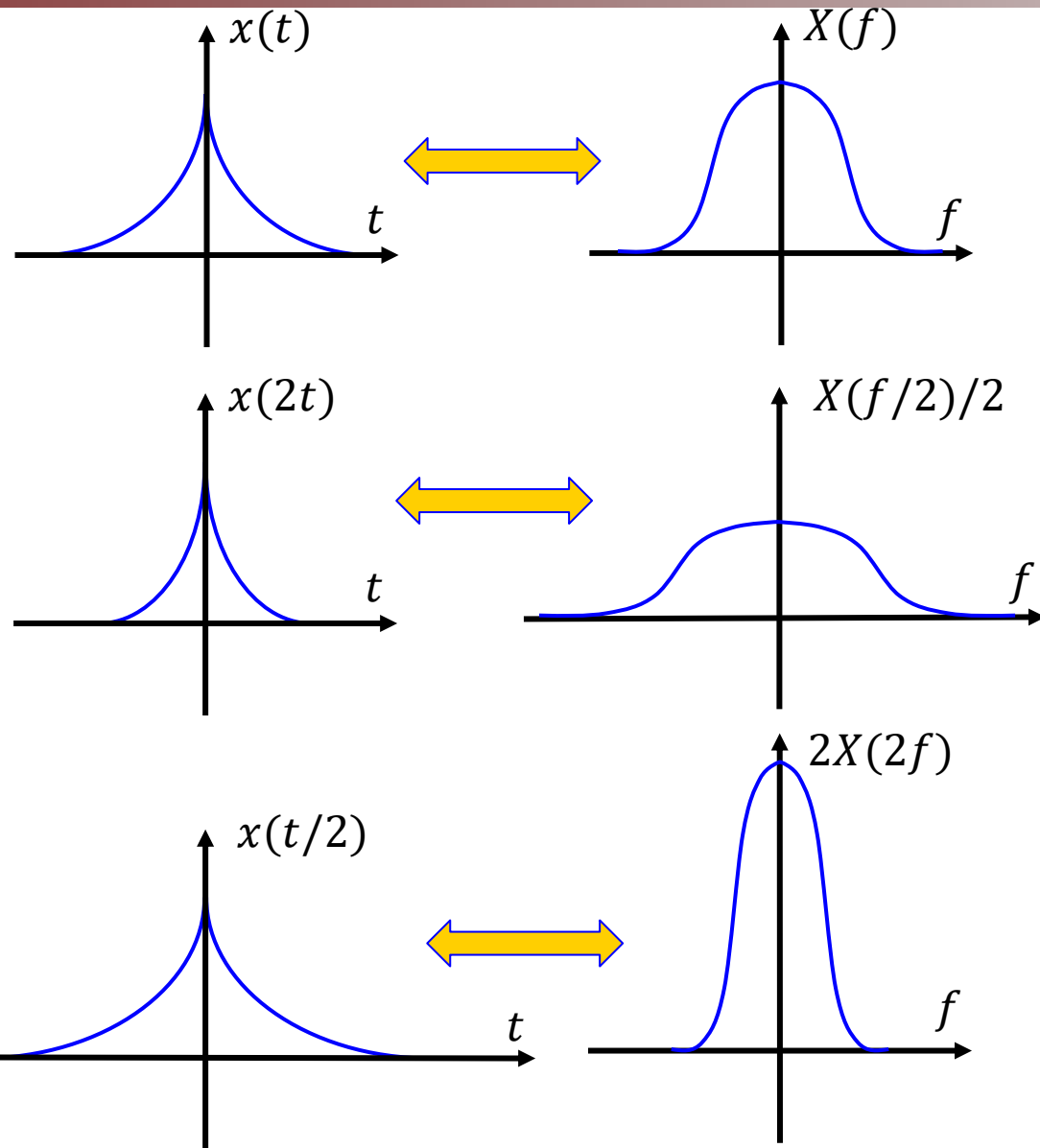
Consider the time-scaling property of Fourier Transforms:

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{f}{a}\right).$$

Suppose $x(t)$ has a bandwidth of B Hz so that $X(f) = 0$ for all $|f| > B$.

Then, $X\left(\frac{f}{a}\right) = 0$ for all $\left|\frac{f}{a}\right| > B$ or equivalently, for all $|f| > aB$. Therefore, $x(at)$ will have a bandwidth of aB Hz.

- ✓ If $a > 1$, compression in time
↔ expansion in frequency,
- ✓ If $a < 1$, expansion in time
↔ compression in frequency.



Time and Bandlimited Signals



- Suppose a signal is time-limited so that $x(t) \neq 0$ only for some range $t_1 < t < t_2$. Furthermore, suppose the signal is bandlimited to some range $-B < f < B$.
- Next suppose we multiply $x(t)$ by a rectangle that is equal to 1 only over $t_1 < t < t_2$. Then, by the multiplication property of F.T.s

$$x(t) \cdot \text{rect}\left(\frac{t - \frac{t_1 + t_2}{2}}{t_2 - t_1}\right) \leftrightarrow X(f) * \left[(t_2 - t_1) \text{sinc}((t_2 - t_1)f) e^{-j2\pi f \left(\frac{t_1 + t_2}{2}\right)}\right]$$

- In the time domain, the signal is unchanged since we are multiplying by 1 when the signal is non-zero and 0 when the signal is zero.
- In the frequency domain, convolving something with a frequency duration of $2B$ with something that has an infinite frequency duration will produce a result that has an infinite frequency duration.
- The only way this can be logically consistent is if $B = \infty$.

- No signal can be simultaneously time-limited and frequency-limited. If it is time-limited it must have infinite frequency duration, if it is frequency-limited, it must have infinite time duration.
- Since we are mortal beings, we cannot create signals of infinite time duration.
- Strictly speaking, any practical signal we create must be of infinite frequency duration.

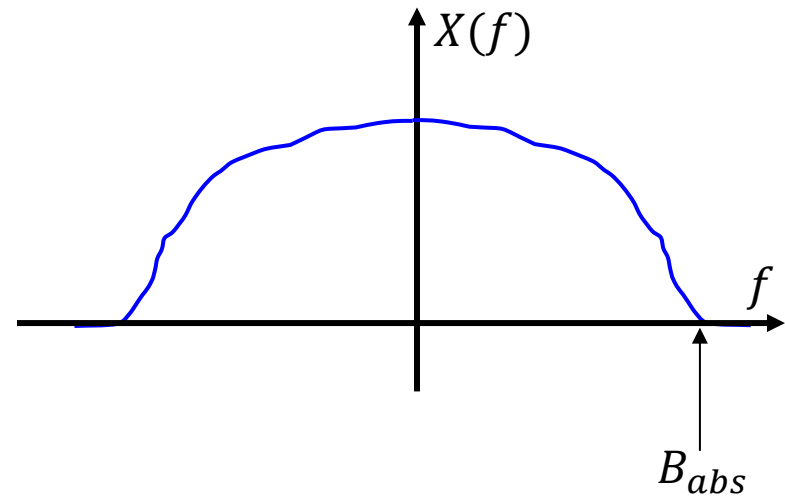
Bandwidth Definitions



- It is common to talk about the bandwidth of a signal which loosely refers to the extent of the signal in the frequency domain.

- The strictest definition of bandwidth measure the full frequency extent of the signal and is called absolute bandwidth:

B_{abs} is the absolute BW of $x(t)$ if B_{abs} is the smallest number such that $|X(f)| = 0$ for all $|f| > B_{abs}$.



- With regard to absolute bandwidth, the previous slide tells us that any signal with finite time duration will have infinite frequency extent.
- If all signals of any practical use have infinite absolute bandwidth, then we need to find a more useable definition of bandwidth.
- Most practical definitions of bandwidth focus on measuring the frequency band that includes “most” of the frequency content.

Power/Energy Bandwidth

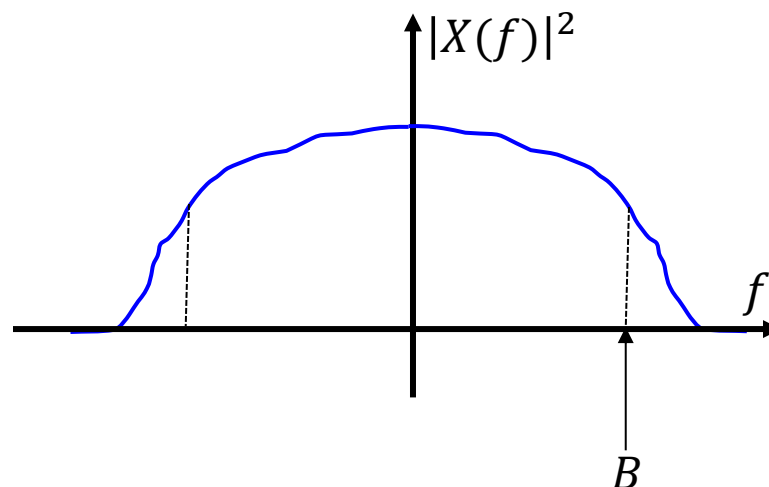


- One “common sense” definition of bandwidth would be to measure the width of the band that contains “most” of the signal power/energy.

- Formally,

P% Energy BW: This is the width of the frequency band which contains P% of the total signal energy.

$$\int_{-B}^B |X(f)|^2 df = \left(\frac{P}{100}\right) \int_{-\infty}^{\infty} |X(f)|^2 df$$



- The energy in the band $-B < f < B$ should be P % of the total energy of the signal.
- Using Parseval's theorem, total energy can be calculated in the time domain if it is more convenient.

$$\int_{-\infty}^{\infty} x^2(t) dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

Caution: You cannot use a similar equation with finite limits. Parseval's theorem only works with infinite limits.

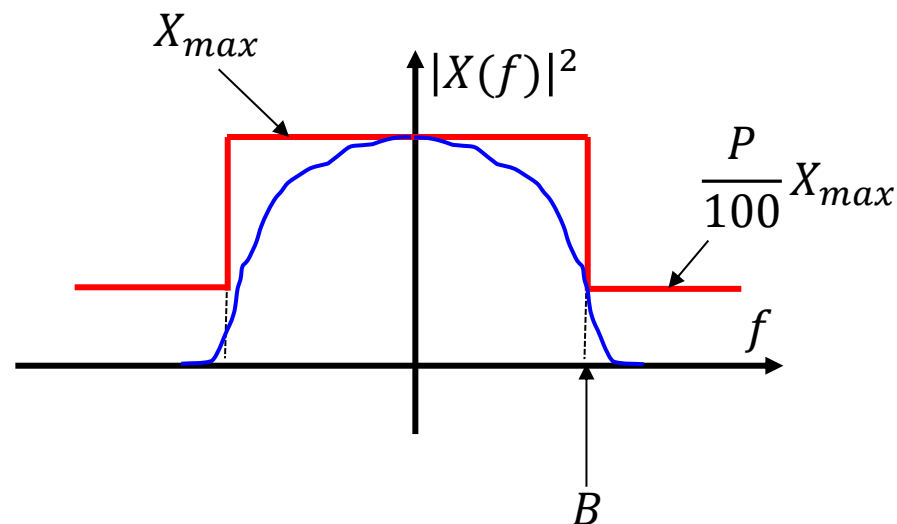
Level Bandwidth



- When the FCC allocates a frequency band for public use, they usually specify a “spectral mask” in which your signal cannot exceed.
- Typical masks are more complicated than the one shown here, but we will use this for illustration purposes.
- In words we might say that the level of the signal must be below a certain threshold outside of our bandwidth.
- Formally, define the P% level bandwidth as:

P% Level BW: B_{lev} is the P% Level BW of $x(t)$ if B_{lev} is the smallest number such that $|X(f)| < \frac{P}{100} X_{max}$ for all $|f| > B_{lev}$.

- Later we will specify the level in decibels rather than as a percentage.



(Noise) Equivalent Bandwidth



- The equivalent bandwidth of a signal is the width of a rectangle that has the same height as the signal and the same total energy.

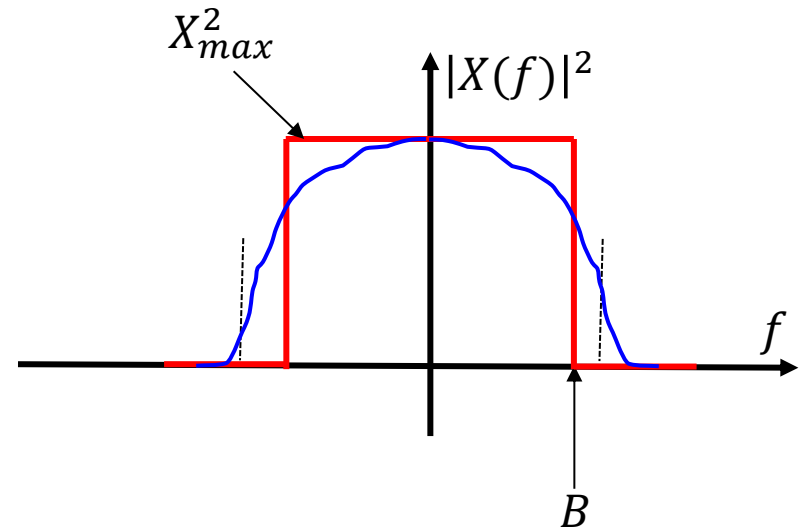
- B_{eq} is chosen so that the area of the rectangle ($= 2B_{eq} X_{max}^2$) is the same as the area under $|X(f)|^2$ ($= \int_{-\infty}^{\infty} |X(f)|^2 df$).

- Therefore

$$\begin{aligned} B_{eq} &= \frac{1}{2X_{max}^2} \int_{-\infty}^{\infty} |X(f)|^2 df \\ &= \frac{1}{X_{max}^2} \int_0^{\infty} |X(f)|^2 df \end{aligned}$$

- Using Parseval's theorem we could also express equivalent BW in terms of an integral in the time domain

$$B_{eq} = \frac{1}{2X_{max}^2} \int_{-\infty}^{\infty} |x(t)|^2 dt$$

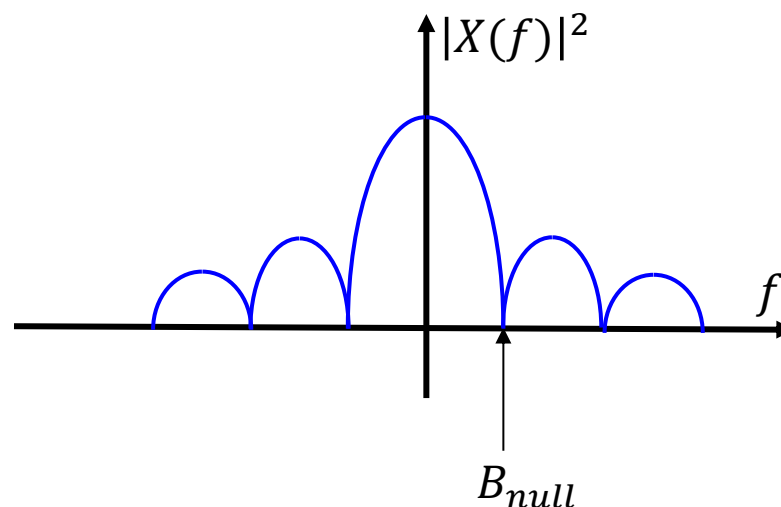


- Equivalent BW is less commonly used than energy or level BW.
- For those familiar with concepts of probability and statistics, equivalent BW measures the width of a signal in the same way that standard deviation measures the width of the distribution of a random variable.

Null Bandwidth



- The null bandwidth of a signal is the lowest (non-zero) frequency at which $|X(f)| = 0$.
- The null BW is of little practical use as there may be significant signal content outside of the null bandwidth.
- The only reason that null BW is sometimes used is because it is often easy to calculate.



Summary of Bandwidth Definitions



- The bandwidth of a signal measures how wide a signal is in the frequency domain. There are several possible ways to measure bandwidth;
 - ✓ Absolute BW: B_{abs} is the absolute BW of $x(t)$ if B_{abs} is the smallest number such that $|X(f)| = 0$ for all $|f| > B_{abs}$.
 - ✓ Null BW: B_{null} is the null BW of $x(t)$ if B_{null} is the smallest number such that $|X(B_{null})| = 0$.
 - ✓ P% Energy BW: This is the width of the frequency band which contains P% of the total signal energy.
 - ✓ Equivalent BW: This will be the bandwidth of a signal with a flat spectrum and same peak spectrum with the same amount of energy as the signal in question. Mathematically it is

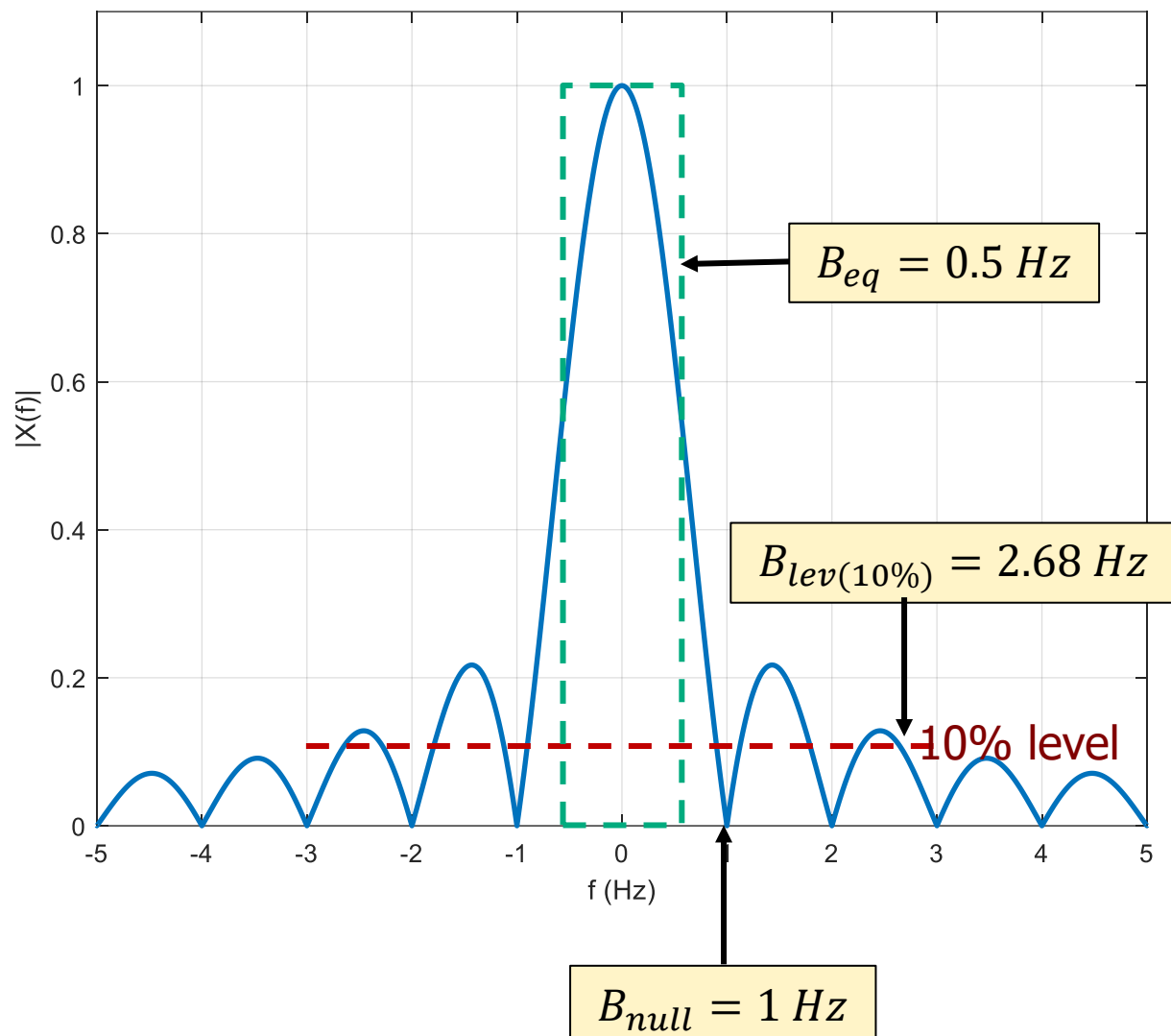
$$B_{eq} = \int_0^\infty \left| \frac{X(f)}{X_{max}} \right|^2 df.$$

- ✓ P% Level BW: B_{lev} is the P% Level BW of $x(t)$ if B_{lev} is the smallest number such that $|X(f)| < \frac{P}{100} X_{max}$ for all $|f| > B_{lev}$.

Bandwidth



Example: $x(t) = \text{rect}(t) \leftrightarrow X(f) = \text{sinc}(f)$, $|X(f)|^2 = \text{sinc}^2(f)$, $X_{\max} = 1$.



$$B_{abs} = \infty$$

$$B_{energy(90\%)} = 0.868 \text{ Hz}$$

$$B_{energy(95\%)} = 2.20 \text{ Hz}$$

$$B_{energy(99\%)} = 11.4 \text{ Hz}$$

Note: When measuring BWs, only measure positive frequencies.

